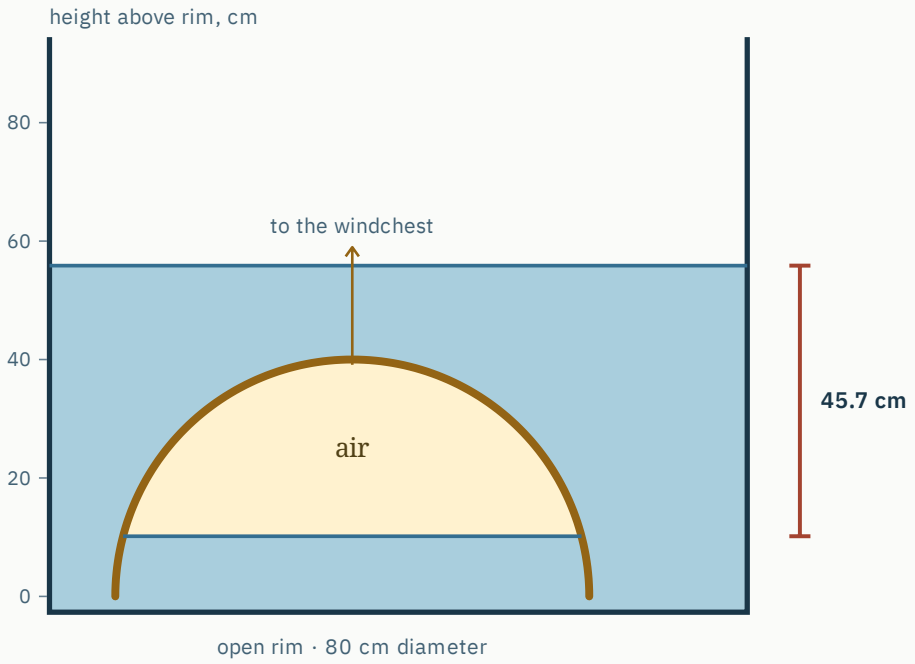
The shape of stored wind

Pnigeus · saved geometric study · hemisphere · 80 cm rim diameter and 40 cm chamber height.

Octagon around the rim circle. Initial water: at the chamber roof. Chosen permitted pressure drop: 30%. This drawing shows the withdrawal endpoint.

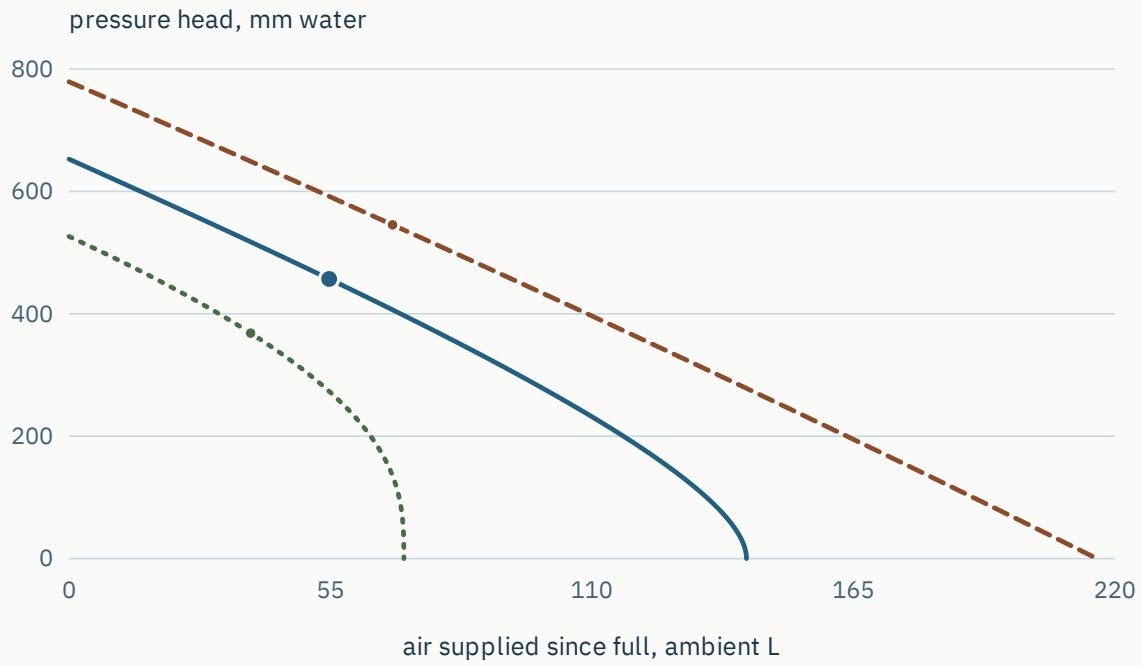
Current gauge pressure 4481.356 Pa; head 456.971 mm; actual air volume 84.071480 L; gas stock 87.789755 ambient-equivalent L. At the withdrawal endpoint, 54.720555 ambient-equivalent L have been supplied and 49.969807 actual L of chamber space have been lost. The chosen pressure endpoint is reached.



The chamber profile and water heights share a scale. Tank width is schematic; the plan choice supplies the actual tank area. Heights start at the open rim; omitted under-rim water is not included.

Same rim and height, same initial fill and tank. Endpoint: 30% below each form's own full pressure.

Chamber form	Full air space actual L	Full head mm water	Air supplied ambient L	Space lost actual L
Hemisphere Hero's stated form	134.04	652.82	54.72	49.97
Cylinder Modern comparison	201.06	779.22	68.05	60.32
Cone Modern comparison	67.02	526.41	38.20	35.89



Solid blue: hemisphere. Dashed brown: cylinder. Dotted green: cone. Dots mark the endpoint in the table. Air supplied includes isothermal compression.

Reopen this exact study. This document embeds its diagrams, complete source page and fonts. Linked external editions need a connection.

Save the exact curves and comparison CSV from this reading. This embedded download also works offline.

Read Hero's Greek

Hero, Pneumatics I.42, Schmidt (1899), p. 192:

ἐν δὲ τῷ ὕδατι κοῖλον ἡμισφαίριον κατεστραμμένον ἔστω

In the water let there be a hollow hemisphere, turned upside down.

New English translation. The transcription rejoins a printed word across lines and removes an older-page marker within κοῖλον.

ἐν δὲ τῷ ὕδατι

en de tōi hydati — In the water. Here ἐν takes the dative for location; τῷ and ὕδατι are dative singular. The headword ὕδωρ changes its stem to ὑδατ- in this form. δέ connects the new instruction to the previous one.

κοῖλον

koilon — Hollow. A neuter singular adjective agreeing with ἡμισφαίριον. A solid half-ball would not hold the air pocket shown in the experiment. The identical neuter nominative and accusative forms are distinguished here by the sentence's construction.

ἡμισφαίριον

hēmisphairion — Hemisphere. A neuter singular noun, the subject of ἔστω. The beginning ἡμι- means half; compare English hemisphere. It tells us a form, but gives no radius, wall thickness or exact water fill.

κατεστραμμένον

katestrammenon — Turned upside down. A perfect middle/passive participle of καταστρέφω, here describing the resulting inverted state. Its neuter singular ending agrees with ἡμισφαίριον, just as κοῖλον does. The printed word continues across two lines.

ἔστω

estō — Let there be. Third-person singular present imperative of εἰμί, to be. Technical Greek often sets up an object this way. The instruction gives us a hollow hemisphere in an inverted position.

Vitruvius's "inverted funnel" need not mean a cone: Roman funnels could be rounded. Hagel (2023, pp. 55–60) discusses the two descriptions together. The cone in this study is a mathematical comparison. Read the research paper.

XLII.

227[29] Ὑδραυλικοῦ ὀργάνου κατασκευή.

Ἔστω τις βωμίσκος χάλκεος ὁ *ΑΒΓΔ*, ἐν ᾧ ὕδωρ
228 ἔστω· ἐν δὲ τῷ ὕδατι κοῖλον ἡμισφαίριον κατεστραμμένον ἔστω, ὃ καλεῖται πνιγεὺς ὁ *ΕΖΗΘ* ἔχων ἐν τῷ ὑγρῷ διάρρυσιν εἰς τὰ πρὸς τῷ πυθμένι μέρη. ἀπὸ δὲ τῆς κορυφῆς αὐτοῦ δύο ἀνατεινέτωσαν σωλῆνες συντετρημένοι αὐτῷ ὑπὲρ τὸν βωμίσκον, εἷς μὲν ὁ *ΗΚΛΜ* κατακεκαμμένος εἰς τὸ ἐκτὸς τοῦ βωμίσκου μέρος καὶ συντετρημένος πυξίδι τῇ *ΝΞΟΠ* κάτω τὸ στόμα ἐχούσῃ καὶ τὴν ἐντὸς ἐπιφάνειαν ὀρθὴν πρὸς ἐμβολέα ἀπειργασμένην. ταύτῃ δὲ ἐμβολεὺς ἁρμοστὸς ἔστω ὁ *ΡΣ*, ὥστε ἀέρα μὴ παραπνεῖν· τῷ δὲ ἐμβολεῖ συμφυὴς ἔστω κανὼν ὁ *ΤΥ* ἰσχυρὸς σφόδρα· πρὸς δὲ τὸν ἁρμόζοντα ἕτερος κανὼν ὁ *ΥΦ* περὶ περόνην κινούμενος τὴν πρὸς τῷ *Υ*· ὁ αὐτὸς δὲ κηλωνευέσθω πρὸς ὄρθιον κανόνα τὸν *ΨΧ* βεβηκότα ἀσφαλῶς. τῇ δὲ *ΝΞΟΠ* πυξίδι ἐπικείσθω κατὰ τὸν πυθμένα ἕτερον πυξίδιον τὸ *Ω* συντετρημένον αὐτῇ καὶ ἐπιπεπωμασμένον ἐκ τῶν ἄνω μερῶν καὶ ἔχον τρύπημα, δι' οὗ ὁ ἀὴρ εἰσελεύσεται εἰς τὴν πυξίδα. ὑπὸ δὲ τὸ τρύπημα λεπίδιον ἔστω ἐπιφράσσον αὐτὸ καὶ ἀνεχόμενον

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1 Hoc caput edidit J. G. Schneider *Eclog. phys.* I, 2, 227—230

5 ἔχ^{o}ων T 7 ἀνατεινέσθωσαν T 15 ἁρμόζοντα **a**: *ad hanc autem accommodata* L: f. πρὸς δὲ τὸν ⟨*ΤΥ* κανόνα ἔστω⟩ ἁρμόζων περὶ A_1 G**b**: παρὰ A_2 T 17—18 τῇ δὲ PT: τῇ A: τὴν G 19 αὐτῇ CG_2P: αὐτῇ δὲ AG_1T 20 ἔχον AG**b**: ἔχ^{o}ων T 22 ἀνεχόμενον PT: ἀντεχόμενον ACG

---

3 τις βωμίσκος χάλκεος ὁ **a**: ἀγγεῖον χάλκεον τὸ **b** 5 ὁ **a**: τὸ **b** ἔχον P 8 ὑπὲρ τὸν βωμίσκον **a**: ὑπερέχοντες εἰς τὰ

Wilhelm Schmidt (ed.), Herons von Alexandria Druckwerke und Automatentheater, vol. I, Teubner, 1899, p. 192. Public-domain edition; scan supplied by the Internet Archive. Inspect the digitized edition.

THE CALCULATION

Water conserved. Air accounted for.

The full chamber starts with inside water at its open rim. Withdrawing air lets that surface rise and the outside surface fall. The vertical difference supports the pressure. The comparison holds the rim radius, chamber height, tank plan and initial fill fixed; the three chambers have different volumes and full pressures.

▼ Read the equations and limits

Heights are measured above the open rim. Let T be the tank area, F(z) the chamber volume below height z, and z_0 the initial water level before pumping. With negligible wall thickness:

```
T × outside − F(outside) + F(inside) = T × z₀
head = outside − inside
gauge pressure = 1000 × 9.80665 × head
ambient gas stock = actual air volume × (1 + gauge pressure / 101325)
```

Volume integrals and monotone bisection solve the two water levels. Gas supplied is the decrease in ambient-equivalent stock. The separate decrease in actual volume is a geometric quantity; compression makes them differ.

The model is isothermal and static. It omits waves, metal thickness, neck and windchest volume, leakage, under-rim clearance, time and pipe acoustics. A high-filled chamber can empty while its last pocket still has a positive limiting head; an empty chamber supplies no wind. The selected pressure drop is a criterion for this experiment, not an ancient performance specification.

Hagel's 80 cm hemisphere example gives about 134 L full volume and 653 mm head. Its roughly 50 L geometric release for a 30% pressure drop becomes about 54.72 ambient-equivalent L under this model's isothermal correction. His paper labels the geometric treatment's omission of compression (p. 65, note 21). These are reconstruction calculations, not measured ancient operating values.

Full derivation and fixed parameters · Sources and rights.

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